Ling 285: Human Language and Technology
Fall 2026

Professor: Khalil Iskarous kiskarou@usc.edu
Office: GFS 301 U
Office Hours: Mon 12:00-1:00pm
Th 1:00-1:50pm, and *by Appointment*
TAs: Muxuan He muxuanhe@usc.edu
Kyle Ng kgng@usc.edu
Zhi Zhang zzheng18@usc.edu

Course Description: Speech is a complex human skill that relies for its functioning on several interlocked physical systems, like the elastic system of tongue movement, the acoustic vibrations of the air in the vocal tract, the marvelous system of sound generation in the larynx, and the phonological structure of a language. Speech science, which is a physical and cognitive science, is now developing at a time when Artificial Intelligence systems that emulate conversation are getting closer and closer to human performance. This class will teach you about how your speech system works, and how technologies that emulate human speech work. The focus will be on how science and technology interact in a loop, so the two will be developed in parallel, and common ideas will be discussed throughout the semester. Physics is a fountainhead of algorithms in machine learning, so physical ideas, their application to speech science, and their algorithmic repercussions will be at the center of the course. The Python language will be used throughout the course to implement ideas, but no prior exposure is necessary.

Course Requirement and grades:

Participation/Attendance/Pop Quizzes:	9%
Exam 1:	25%
Exam 2:	25%
Final:	25%
HW:	14%
Experiment Participation:	2%

Please Always bring a device to class that you can use for online pop quizzes

HW: There will be 9 HWs in the semester, and you will always have 7 days to complete them. Students can help each other work on the material, but each student must do the HW separately. No use of AI for HW. **The lowest HW grade will be dropped.**

Lab: Lab is *mandatory* and will exercise material discussed in the course.

Academic Integrity: For all of your written work, quotations or other forms of reference to other work are of course acceptable with the appropriate citations. USC takes academic integrity very seriously and enforces it strictly.

Experiment Participation: 2% of the grade will be for participation in psycholinguistic experiments approved by the Institutional Review Board at USC. Such participation will enable you to understand how such experiments are conducted. If you wish to not participate, there will be an assignment in November whose completion requires about the same amount of time as experiment participation.

Academic Integrity

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct is contrary to this fundamental mission and includes any act of dishonesty in the submission of academic work (either in draft or final form), as well as cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage. Students are expected to uphold the highest standards of academic integrity in all coursework.

This course follows the expectations for academic integrity as stated in the USC Student Handbook. All students are expected to submit assignments that are original work and prepared specifically for the course/section in this academic term. Students may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of academic misconduct will be reported to the Office of Academic Integrity.

Academic dishonesty has a far-reaching impact and is considered a serious offense against the university. Violations will result in a grade penalty, such as a failing grade on the assignment or in the course, and disciplinary action from the university, such as suspension or expulsion.

For more information about academic integrity see the Student Handbook, the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

[In this course, unless specifically designated as a 'group project,' **all assignments are expected to be completed individually.** In this class, you are expected to submit work that demonstrates your individual mastery of the course concepts.] Please ask me if you are unsure about what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Course Content & Recording Distribution Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Violations of Academic Integrity: Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an OSAS accommodation. Recording can inhibit free discussion, and thus infringe on the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 16).

Violations of Academic Integrity: Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposes other than individual or group study. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which had been distributed to students or in any way had been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. Distributing course material without the instructor's permission will be presumed to be an intentional act to facilitate or enable academic dishonestly and is strictly prohibited. (Living our Unifying Values: The USC Student Handbook, page 16).

In this course, such violations include the submission to AI of notes provided.

Course Evaluations

Course evaluation by students occurs at the end of the semester university-wide.

Accommodations

OSAS is responsible for the determination of appropriate accommodations. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Counseling and Mental Health - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices.

CARE-SC: Confidential Advocacy, Resources, and Education Support Center - (213) 740-9355(WELL) –24/7/365 on call.

Confidential advocates, prevention educators, and professional counseling teams work to promote a universal culture of consent, as well as prevent and respond to sexual assault, intimate partner violence, stalking, or other relationship harm. Services available to all USC students at no cost.

Office of Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment, discrimination, retaliation on the basis of a protected characteristic, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

USC Campus Support and Intervention - (213) 740-0411

Focuses on student success by assisting students in navigating and resolving complex issues through problem solving, presenting options, and connecting to resources

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety -

For 24 hour emergency assistance or to report a crime: UPC: (213) 740-4321, HSC: (323)-442-1000.

For 24 hour non-emergency assistance or information: UPC: (213) 740-6000, HSC: 323-442-1200.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

The Office of Sustainability promotes numerous ways for all Trojans to study, work, and live sustainably at USC. Students can access tools, resources, and opportunities on their website and in the Sustainability Hub in the Student Union Building, Suite 101.

USC Report & Response provides a front door to reporting concerns, including an ability to report anonymously, which violate university policies or go against our Unifying Values and helps connect you with the support you need. The website provides a list of confidential resources, supporting offices, and policies that are available to our university community. The Resources page includes information regarding the university's response in the event of immigration

enforcement activity on campus or involving a student or employee. Additional questions may be directed to the Office of Professionalism and Ethics (OPE) and ope@usc.edu or in reviewing the Office of Professionalism and Ethics website. (213) 740-2500

Schedule	Topic
August 25 August 27	Speech Segments 1, Stochastic Gradient Descent Speech Segments 2, Dynamic Programing (Reading: Meadows)
Sept 1 Sept 3	Speech Segments 3, Dynamical Systems 1 (Reading: Meadows) Speech Segments 4, Dynamical Systems 2 (Reading: Meadows)
Sept 8 Sept 10	Syllables 1, Memory and Outer Products (Reading: Meadows) Syllables 2, Analysis (Reading: Meadows) **HW1**
Sept 15 Sept 17	Masses and Springs 1, Chain Rule (Reading: Handout) Masses and Springs 2, Some Animal Neuroscience (Reading: Handout) **HW2**
Sept 22 Sept 24	Neural Networks, Human and Artificial 1 Neural Networks, Human and Artificial 2 **HW3**
Sept 29 Oct 1	Review **Exam 1**
Oct 6 Oct 8	Neural Networks 3 **No Class, Fall Recess**
Oct 13 Oct 15	LLM Structure 1, More Acoustics LLM Structure 2 **HW4**
Oct 20 Oct 22	LLM Structure 3 LLM Structure 4 **HW5**
Oct 27 Oct 29	A little Kant and Hegel LLM Learning 1 **HW6**
Nov 3 Nov 5	LLM Learning 2, Speech Recognition System 1 **HW7**
Nov 10 Nov 12	Speech Recognition System 2 Speech Recognition System 3 **Exam 2**
Nov 17 Nov 19	Duplex Conversation 1 Duplex Conversation 2 **HW8**
Nov 24 Nov 26	Review **No Class, Thanksgiving**
Dec 1 Dec 3	Voice Assistants Review **HW9**
Final Exam	Thursday, December 10, 11:00am-1:00pm